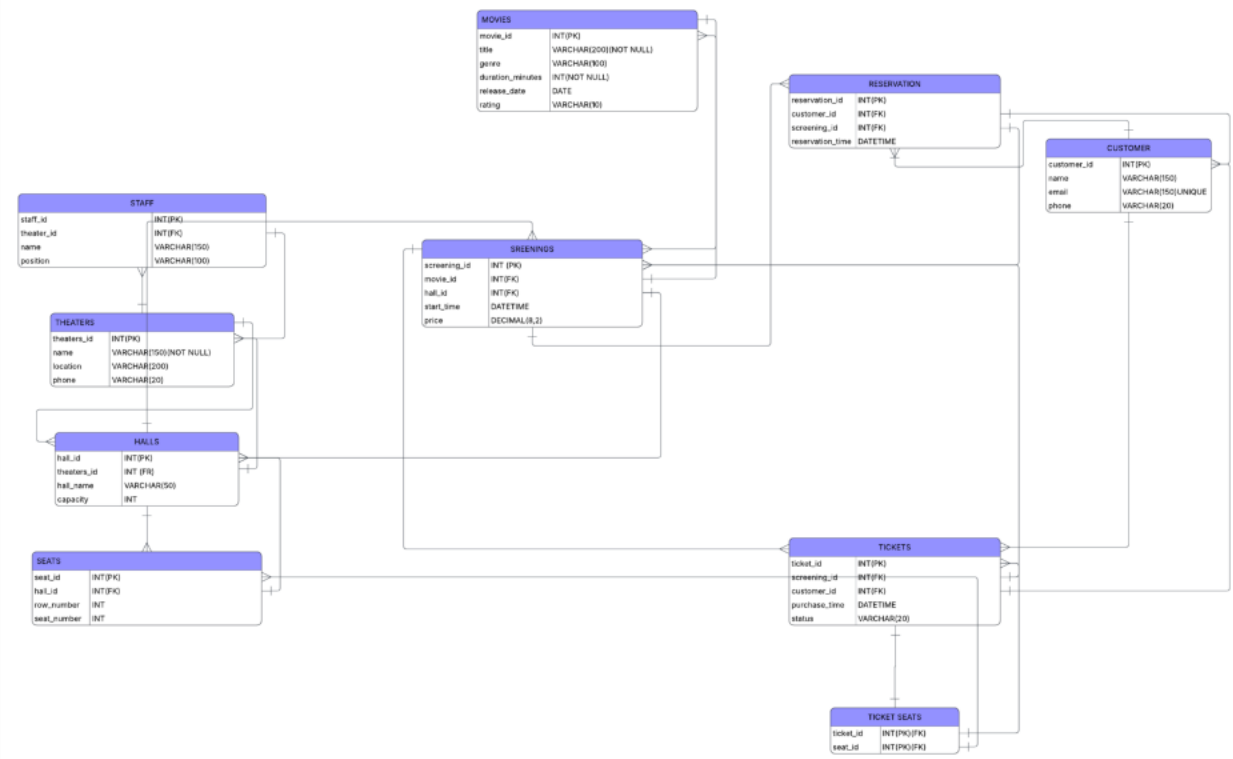
## 1.1 Logical Scheme




## 1.2 Objects

### Table Description (Movies)

The table stores information about movies shown in the movie theater. It contains basic details such as the movie title, genre, duration, and release date.

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| MOVIES | movie_id | Unique identifier for each movie | INT, PK, Auto increment |

| | | | |
|---|---|---|---|
| | Title | Name of the movie | VARCHAR(200), NOTNULL |
| | Genre | Movie genre | VARCHAR(100) |
| | duration_minutes | Duration of the movie minutes | INT |
| | release_date | Date the movie was released | DATE |
| | Rating | Age rating of the movie | VARCHAR(10) |

Comments on table relationships

One to Many relationship with **Screenings** (one movie can have multiple screenings)

| movie_id | title | duration_minutes | rating |
|---|---|---|---|
| 1 | Avatar | 162 | PG-13 |

Table Description (Theaters)

The table stores information about movie theaters where films are screened

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| Theaters | theater_id | Unique identifier for each theater | INT, PK, Auto increment |

| | | | |
|---|---|---|---|
| | name | Name of the movie | VARCHAR(150), NOTNULL |
| | location | Movie genre | VARCHAR(200) |
| | phone | Duration of the movie minutes | VARCHAR(20) |

## Comments on table relationships

One-to-Many relationship with Halls

One-to-Many relationship with Staff

| theater_id | name | location |
|---|---|---|
| 1 | Cinema City | Downtown |

Table Description (Halls)

This table stores information about halls inside each theater where movies are shown

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| HALLS | hall_id | Unique identifier for each hall | INT, PK |

| | | | |
|---|---|---|---|
| | theater_id | Reference to the theater | INT,FK |
| | hall_name | Name of the hall | VARCHAR(50) |
| | capacity | Maximum number of seats | INT |

Comments on table relationships

Many-to-One with Theaters

One-to-Many with Seats

One-to-Many with Screenings

| Hall_id | theater_id | hall_name | capacity |
|---|---|---|---|
| 1 | 1 | Hall 1 | 120 |

Table Description (Seats)

This table stores information about seats in each hall

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| SEATS | seat_id | Unique identifier for each seat | INT,PK |

| | | | |
|---|---|---|---|
| | hall_id | Reference to the hall | INT,FK |
| | row_number | Row number of the seat | INT |
| | seat_number | Seat number | INT |

## Comments on table relationships

Many-to-One with Halls

Many-to-Many with Tickets through Ticket_Seats

| Seat_id | hall_id | row_number | Seat_number |
|---|---|---|---|
| 1 | 1 | 5 | 8 |

Table Description (Screenings)

This table schedules movies in specific halls at a certain time

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| SCREENING | screening_id | Unique identifier of the screening | INT,PK |
| | movie_id | Reference to the movie | INT,FK |

| | | | |
|---|---|---|---|
| | hall_id | Reference to the hall | INT,FK |
| | start_time | Start time of the screening | DATETIME |
| | price | Ticket price | DECIMAL(8,2) |

## Comments on table relationships

Many-to-One with Movies

Many-to-One with Halls

One-to-Many with Tickets

| screening_id | movie_id | hall_id | Start_time |
|---|---|---|---|
| 1 | 1 | 1 | 2024-05-10 18:00 |

Table Description (Customers)

The table stores information about customers who purchase or reserve tickets

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| CUSTOMER | customer_id | Unique identifier for each customer | INT,PK |
| | name | Customer name | VARCHAR(150) |

| | | | |
|---|---|---|---|
| | email | Customer email | VARCHAR(150) |
| | phone | Phone number | VARCHAR(20) |

## Comments on table relationships

One-to-Many with Tickets

One-to-Many with Reservations

| Customer_id | name | email |
|---|---|---|
| 1 | John Smith | john@email.com |

Table Description (Ticket Seats)

This table implements the Many-to-Many relationship between Tickets and Seats

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| TICKET SEATS | ticket_id | Reference to the ticket | INT, PK, FK |
| | seat_id | Reference to the seat | INT, PK, FK |

## Comments on table relationships

Many-to-One with Tickets

Many-to-One with Seat

| Ticket_id | seat_id |
|---|---|
| 101 | 15 |

Table Description (Tickets)

This table stores information about purchased tickets for movie screenings. It records which customer bought the ticket and for which screening

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| TICKETS | ticket_id | Unique identifier for each ticket | INT, PK, Auto increment |
| | screening_id | Reference to the screening | INT, NOT NULL, FK |
| | customer_id | Reference to the customer who purchased the ticket | INT, NOT NULL, FK |
| | purchase_time | Time when the ticket was purchased | DATETIME |
| | status | Ticket status (paid, cancelled, reserved) | VARCHAR(20) |

## Comments on table relationships

Many-to-One relationship with Customers (one customer can buy multiple tickets)

Many-to-One relationship with Screenings (one screening can have many tickets)

Many-to-Many relationship with Seats through the Ticket Seats table

| **ticket_id** | screening_id | customer_id | purchase_time |
|---|---|---|---|
| 101 | 1 | 1 | 2024-05-10 17:45 |

Table Description (Reservations)

This table stores information about seat reservations made by customers before purchasing tickets

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| RESERVATIONS | reservation_id | Unique identifier for each reservation | INT, PK |
| | customer_id | Reference to the customer | INT, NOT NULL, FK |
| | screening_id | Reference to the screening | INT, NOT NULL, FK |
| | reservation_time | Time when the reservation was made | DATETIME |

## Comments on table relationships

Many-to-One relationship with Customers (one customer can make multiple reservations)

Many-to-One relationship with Screenings (one screening can have many reservations)

| reservation_id | customer_id | screening_id | reservation_time |
|---|---|---|---|
| 1 | 1 | 1 | 2024-05-10 16:30 |

Table Description (Staff)

This table stores information about employees working at the movie theater

| Table Name | Field name | Field Description | Data Type / Constraints |
|---|---|---|---|
| Table 1 | Staff_id | Unique identifier for each employee | INT, PK |
| | theater_id | Reference to the theater where the employee works | INT, NOT NULL, FK |
| | name | Employee name | VARCHAR(150), NOT NULL |
| | position | Job position (manager, cashier, technician) | VARCHAR(100) |

## Comments on table relationships

Many-to-One relationship with Theaters (one theater can employ many staff members)

| **staff_id** | theater_id | name | position |
|---|---|---|---|
| 1 | 1 | Anna Brown | Manager |